

2019 Republic of China Biology Olympiad National Team

Selection Camp

Theoretical Assessment Test Questions

Part One

Student Code



Precautions:

1. This section of the test consists of 12 pages (including the cover) and is worth 100 points.
2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please make sure that the cover of the exam paper matches your exam number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part 1: (100 points)

I. International Competition Type Questions (25 points total) Note :

International competition type questions are graded at unequal intervals. Each question set contains four sub-questions. A score of 0 points is awarded for answering all four sub-questions incorrectly.

You will receive 0.5, 1, 2.5 and 5 points for answering questions 1, 2, 3 and 4 correctly, respectively .

Report 1:

James Allison and Tasuku Honjo shared the 2018 Nobel Prize in Medicine for their discovery that different strategies of inhibiting the immune system's "brakes" can be used to fight cancer, significantly improving the treatment outcomes for several previously difficult-to-treat cancers. These proteins, acting as the brakes on the immune system, hinder the effectiveness of T cells in attacking cancer cells. Allison studied the cytotoxic T-cell antigen-4 (CTLA-4), finding that blocking this protein would release the "brakes," allowing immune cells to attack tumors. Honjo, on the other hand, discovered a...

A novel protein, the T-cell inhibitory receptor PD-1, has been identified as a suppressive immune checkpoint. Following the initiation of T-cell activity, PD-1 expression increases and it binds to PD-L1/PD-L2, thereby disrupting T-cell antigen receptors.

The body intervenes in signal transduction, reduces proliferation, inhibits cytokine production, and promotes apoptosis, thereby suppressing T cell immunity.

The immune response. Their discovery pioneered a new treatment for cancer, namely immune checkpoint inhibition therapy, for malignant melanoma. This provides new options for patients with melanoma, lung cancer, and bladder cancer.

Determine whether the following four statements are true or false, and fill in **T** (true) or **F** (false) in the corresponding column of the table below.

(A) Activation of immune cells often requires stimulation with two signals, the second of which is called costimulation.

(costimulatory) signals can be generated by the binding of costimulatory molecules on the surface of antigen-presenting cells to costimulatory molecule receptors (such as CD28) on the surface of T cells.

(B) T cells also have a CTLA-4 molecule on their surface, which competitively binds to CD28, generating a second signal upon binding. If a monoclonal antibody against CTLA-4 is prepared, it can bind to the CTLA-4 molecule, inhibiting the second signal.

produce.

(C) PD-1 can be used as a target for immune checkpoint inhibitor therapy; anti-PD-1 monoclonal antibodies can fully activate T cells.

Cells, preventing T cell line loss and dysfunction, and developing new drugs to fight cancer.

(D) Immune checkpoint inhibitors are different from conventional chemotherapy drugs. They do not have direct cytotoxicity to tumor cells, but by blocking the tumor's suppression of the immune system, they can enhance the patient's own immune response against the tumor.

Narrative	(A)	(B)	(C)	(D)
T or F				

Report 2:

African swine fever first broke out in Kenya in 1921, approximately 100 years ago.

Currently, 47 countries have been designated as epidemic areas, mostly in Europe and Africa. The outbreak was first reported in mainland China last year and quickly spread.

African swine fever has spread, becoming a major epidemic area in Asia. Taiwan activated a national alert, strictly prohibiting the import of pork from infected areas, implementing disinfection

measures at livestock farms, and managing the safe use of kitchen waste in pig feed to prevent outbreaks and subsequent designation as an epidemic area, the economic impact of which

would be incalculable. In March 2019, Taiwan specially invited experts from Eurasian countries to participate in the "International Symposium on Prevention and Control of African Swine

Fever" to share their experience in African swine fever prevention and control and to exchange technical information. The experts at the symposium agreed that transparent reporting, accurate

and rapid diagnostic techniques, strengthened border control, prevention of smuggling, and raising public awareness of disease prevention are crucial to preventing the spread of African swine fever.

An important means of spreading the epidemic in Asia.

Determine whether the following four statements are true or false, and fill in **T** (true) or **F** (false) in the corresponding column of the table below.

(A) A dead pig was found washed ashore on the southwest coast of Lieyu, Kinmen County. A veterinarian isolated a pathogen from the dead pig.

Based on the three characteristics of the pathogen: (1) it does not have flagella, (2) it contains DNA and protein, and (3) it can multiply in an environment containing multiple antibiotics, it can be determined that this pathogen is a virus.

(B) African swine fever is highly contagious. Infected pigs will develop high fever, purplish spots on their skin, and internal bleeding. The mortality rate of infected pigs can reach 100%. Currently, there are no drugs available for treatment, nor are there suitable treatments.

We use vaccines for prevention.

(C) After nearly 100 years of research, it is known that the pathogen of African swine fever is a virus, like all viruses that cause swine fever, with double-stranded DNA (dsDNA), and belongs to the same family of swine fever virus (Flaviviridae).

(D) To date, no cases of human infection with African swine fever virus have been found, indicating species specificity of the host. Furthermore, because it is a DNA virus, the chance of mutation is very small, and in the past nearly 100 years, no cases have emerged.

African swine fever has been transmitted across species due to mutations, so theoretically, it will be eradicated when all pigs infected with the African swine fever virus die out in all pig farms, whether in Europe or Africa.

Narrative	(A)	(B)	(C)	(D)
T or F				

Report 3:

Peter Doherty has long been concerned with human health issues and has recently advocated for a citizen science approach.

To protect the health of both humans and birds, he has authored a new book, "The Fate of Birds is the Fate of Humans," which points out that birds are a reflection of human well-being.

A health and early warning system for the Earth's ecology is like how miners used to bring canaries into the mines, because canaries...

Canaries are extremely sensitive to toxic gases; even a slight change in the surrounding air will cause them to immediately cry out or even die. The book also discusses...

Viruses that seriously affect bird health, such as West Nile virus and avian influenza virus, and an introduction to bird diseases, such as...

Psittacosis, malaria, cancer. For birds, humans may be an even more terrifying pathogen, because humans not only create...

Pollution, and overhunting. But Duhti's most famous research is in immunology; he discovered the killer T cell...

Cytotoxic T cells only kill major histocompatibility complexes.

Infected cells with MHC complexes, the interaction between the two cell types is limited by histocompatibility antigens, and also because...

This research earned him the Nobel Prize, the same one awarded to Rolf Zinkernagel.

Determine whether the following four statements are true or false, and fill in **T** (true) or **F** (false) in the corresponding column of the table below.

(A) West Nile fever, avian influenza, and psittacosis are all infectious diseases shared by humans and birds; the pathogens are all viruses.

They all originated in Africa.

(B) MHC is related to tissue compatibility during organ transplantation and also to the host's fight against viruses. T cells utilize T...

For cell receptors (TCRs) to recognize viral antigens, MHC molecule recognition is also required to kill the receptors.

Cells infected by a virus.

(C) Self recognition, antigen presentation, T-cell receptor recognition, thymocyte positive selection, and antibody class switching.

(Ab class switch) are all related to MHC restriction.

(D) Human MHC, also known as human leukocyte antigen (HLA), is located in [location missing].

The short arm of chromosome 6, composed of 3.6 million base pairs, is currently the densest known gene-producing chromosome in the human chromosome.

The region with the highest degree of polymorphism can serve as an in vivo molecular fingerprint.

Narrative	(A)	(B)	(C)	(D)
T or F				

Report 4:

The Centers for Disease Control and Prevention (CDC) specifically reminds the public that the influenza virus remains active. According to monitoring data, the 2019 influenza outbreak in Taiwan...

In mid-March, the cumulative number of outpatient and emergency room visits for influenza-like illnesses reached 82,252, a slight decrease of 5.7% compared to the beginning of March. Last week, 55 new cases of severe influenza complications were reported, primarily caused by influenza A, with H1N1 accounting for 78% (43 cases), and 65 of these cases being H1N1.

Of the 28 cases (51%), the majority were over 60 years old; the remaining 7 cases, upon review, were all found to be deaths caused by influenza A.

The cases, aged between 34 and 86 years, all had a history of chronic diseases and had not been vaccinated against influenza. Since October 2018, China has recorded a cumulative total of 716 cases of severe influenza complications (440 cases infected with H1N1 and 242 cases infected with H3N2).

(Multiple cases), including 70 deaths (mostly 41 cases of H1N1 infection and 25 cases of H3N2 infection). In neighboring regions, except for southern mainland China and Hong Kong where H1N1 remains dominant, most other countries have seen a co-circulation of H1N1 and H3N2, with Japan and South Korea primarily experiencing H3N2 transmission. The most serious infectious disease in human history was...

The Spanish flu of 1918 spread throughout the world within a year, killing more than 50 million people by the end of that year, exceeding the total number of deaths in World War I. It has now been exactly 100 years since then.

Determine whether the following four statements are true or false, and fill in T (true) or F (false) in the corresponding column of the table below.

- (A) Influenza virus is a negative-sense single-stranded RNA (ssRNA) virus with a mantle. It belongs to the Paramyxoviridae family and can be distinguished into types A, B, and C using the M1 protein. In terms of type antigen characteristics, type C virus is the most stable and does not mutate easily, while type A virus is the most prone to antigenic mutation.
- (B) Influenza A virus has 8 gene segments encoding PB1, PB2, PA, HA, NP, NA, M1, M2, NS1, NS2. There are 10 proteins, including NS2. Among them, HA (hemagglutinin) and NA (neuraminidase) are glycoproteins on the mantle surface and are the basis for distinguishing viral strain subtypes such as H1N1 or H5N1. They are prone to antigenic mutations, including antigenic drift caused by point mutations and antigenic shift caused by gene recombination.
- (C) HA is a crucial factor determining whether a virus can enter a host cell. After hydrolysis, it separates into two parts: a light chain and a heavy chain. The light chain can bind to sialic acid receptors on the host cell membrane, and the light chain can assist in the fusion of the viral mantle with the host cell membrane. When the replicated virus wants to leave the host cell by budding, the HA on the viral surface maintains a connection with the host cell through sialic acid, while NA can hydrolyze sialic acid to sever the connection between the host cell and the virus, assisting in viral detachment. Infected cells.
- (D) The main component of currently used influenza vaccines is the NA antigen fragment of the virus, which can cause the immune system to produce antibodies to attack the virus. However, new vaccines are needed when new viruses emerge. Even viruses common during the flu season can mutate slightly. Therefore, vaccine manufacturers often need to anticipate changes and adjust existing formulas to make new vaccines effective against new viruses.

Narrative	(A)	(B)	(C)	(D)
T or F				

Report 5:

Signal transduction is a form of communication between cells, achieved through the interaction of ligands and receptors.

Interactions transmit information from outside the cell to inside, triggering various cellular responses and utilizing signaling pathways.

They interact to form networks. Information transmission is crucial for basic cellular activities and coordinated responses; cells...

The ability to recognize and respond appropriately to the surrounding microenvironment is crucial for individual development, tissue repair, immunity, and homeostasis.

Based on certain fundamental principles, errors in cell information processing can lead to cancer, diabetes, and autoimmune diseases.

Diseases such as...

Determine whether the following four statements are true or false, and fill in **T** (true) or **F** (false) in the corresponding column of the table below.

(A) Receptors located on the cell surface, including G protein-coupled receptors (GPCRs).

The tyrosine protein kinase receptor family, etc., by Robert Rekowitz of the United States.

Lefkowitz and Brian Kobilka previously collaborated on research into the G protein-coupled receptor family (GPCRs), and

Its interaction with ligands earned it the Nobel Prize in Physiology or Medicine.

(B) The causes of cancer include: (1) the action of oncogenes (such as *sis*) , which cause cells to continuously secrete growth factors and grow.

(1) Growth factors act on receptors on their own cell membranes, causing cells to proliferate continuously; (2) They control cell surface growth factor receptors.

Mutations in tumor suppressor genes (such as *erb-B*) can cause receptors to become inactive.

(C) GPCRs constitute a large family of receptors, with ligands including adrenaline, dopamine, serotonin, histamine, etc.

Caffeine and many other commonly used drugs also exert their effects by interacting with GPCRs. Receptor molecules consist of a multi-channel GPCR.

Composed of peptide chains, containing 7 hydrophobic amino acids, it can cross the cell membrane 7 times. Upon stimulation, it first activates the receptor periphery.

G proteins, and then the subunits of G proteins, can activate or inhibit specific proteases or ion channels in the surrounding environment.

road.

(D) Some receptors are located inside the cell, such as nuclear receptors, GPCRs, and ion channel receptors.

Ionotropic receptors, etc., and nuclear receptors are mostly transcription factors, which are stimulated...

It can then bind to specific genes, affecting protein production.

Narrative	(A)	(B)	(C)	(D)
T or F				

II. True or False Questions (4 points **each** , **24 points in total**)

Determine whether the following four statements are true or false, and fill in **T** (true) or **F** (false) in the corresponding column of the table below.

1. In eukaryotes, chromosomes are the genetic material within cells that possesses hereditary properties, formed by the compression of chromatin.

Aggregates are easily stained dark by basic dyes, hence the name chromosome.

(A) Chromosomes exist in pairs in somatic cells, but individually in germ cells.

(B) The nucleosome is the most basic structural unit of a chromosome. It is composed of four pairs of histones (H2A, H2B, H3, and H4).

A flattened spherical tetramer composed of two molecules of H4.

(C) Chromosome identification is represented by letters and numbers, for example, "9q34.1" indicates chromosome 9.

The first subzone of the fourth band in the third region of the short arm of the body.

(D) Chromosomes form before cell division. During the synthesis phase (S phase) of the cell, chromosomes disperse into primary chromosomes.

Deoxyribonucleic acid molecules, whether structural or extended, make up the chromatin structure within the cell nucleus.

Narrative	(A)	(B)	(C)	(D)
T or F				

2. Regarding the following, apoptosis is a type of programmed cell death.

(A) Compared to necrosis, apoptosis is an active response of the cell.

(B) During apoptosis, events such as chromatin condensation occur.

Terminal chromosome fragmentation and nuclear fragmentation, etc.

Phenomenon.

(C) In the body of a typical adult, an average of millions of cells undergo apoptosis every day.

(D) When proteins within a cell are continuously damaged, they rapidly begin to contract, disintegrate, and then form fragments.

Finally, they are cleared away by macrophages through phagocytosis.

Narrative	(A)	(B)	(C)	(D)
T or F				

3. Signal transduction is the process by which chemical or physical signals are transmitted through cells as a series of molecular events.

The most common process of transmission is protein phosphorylation catalyzed by protein kinases.

(A) The protein responsible for detecting stimuli is usually called a receptor, and the protein that binds to the receptor is called a ligand.

(B) The ligand is called the first messenger, while the receptor is the signal sensor, which then activates the primary effector.

(C) Primary effectors are usually associated with second messengers, which can activate secondary effectors.

(D) The process of information transmission between nerve cells, such as acetylcholine, also follows the above pattern to amplify signals.

Narrative	(A)	(B)	(C)	(D)
T or F				

4. Targeted therapy primarily refers to treatments where the medication targets mutated cells (such as cancer cells).

It acts as a target molecule. More accurately, it refers to "molecular targets" on or within cancer cells.

This includes receptors, genes, or signaling pathways related to tumor growth, as well as tumor angiogenesis factors.

(A) Targeted therapy can reverse the malignant phenotype of tumor cells.

(B) Taking cancer treatment as an example, its main target is cancer stem cells.

(C) Immunotherapy (the treatment using immune mechanisms) is a type of targeted therapy.

(D) Tumors are not formed by the continuous accumulation of mutations from a single normal cell line (clone), as previously thought.

It is not a tumor, but a group of cells with genetic differences. Therefore, each target must be...

different.

Narrative	(A)	(B)	(C)	(D)
T or F				

5. In eukaryotic cells, transcription refers to the process by which genetic information is converted from DNA to RNA.

Translation is the first step in the process of protein biosynthesis.

(A) Transcription occurs in the cell nucleus, while translation occurs in the cytoplasm.

(B) Transcription occurs in the cell nucleus, translation occurs in the cytoplasm, and then the transcription is transported back to the nucleus for further processing.

(C) Transcription and translation both occur in the cell nucleus; subsequent transport and modification take place in the cytoplasm.

(D) Transcription and translation occur continuously and without interruption between the cell nucleus and cytoplasm.

Narrative	(A)	(B)	(C)	(D)
T or F				

6. The origin of eukaryotes is an interesting evolutionary puzzle.

(A) Eukaryotes originated from archaea infected with poxviruses.

(B) Proto-eukaryotic cells are directly derived from archaea.

(C) DNA replication errors or polyploidization lead to the formation of a complex internal membrane system.

(D) The symbiotic origin within mitochondria is an attack by one prokaryote on another.

Narrative	(A)	(B)	(C)	(D)
T or F				

III. Multiple Choice Questions (8 points in total)

The following are some common bacteria. Please answer the following multiple-choice questions based on their characteristics. You will only receive points if you answer all questions correctly.

- A. *Bacillus subtilis*
- B. *Escherichia coli*
- C. *Staphylococcus aureus*
- D. *Mycoplasma pneumoniae*
- E. *Pseudomonas aeruginosa*
- F. *Clostridium tetani*
- G. *Streptococcus lactis*

1. Which of the above bacteria can produce endospores? **(2 points)**

2. Which of the above bacteria has flagella? **(2 points)**

3. After Gram staining, which of the above bacteria will show a blue-purple color? **(2 points)**

4. Which of the above bacteria does not have a peptidoglycan cell wall? **(2 points)**

IV. Short Answer Questions (17 points in total)

1. *Dinoflagellates* is a single-celled alga capable of photosynthesis. Please briefly answer the following questions:

(1) Its chloroplasts have 4 membranes, which differs from the double-membrane chloroplasts of common green algae and plants. Please explain why.

The reason. **(2 points)**

(2) When this type of algae proliferates in large quantities, it causes the water to turn red, which is called a red tide. Please briefly describe it.

Explain why it appears red. **(2 points)**

(3) Briefly describe the impact of red tides on marine life. **(2 points)**

2. Regarding purple sulfur bacteria, cyanobacteria, and green algae, please answer the following questions. (1) Compare the similarities and differences in their cellular structures. (1 point)

(2) Please compare the similarities and differences in the cell wall components of the three. (1 point)

(3) What are the similarities and differences in the photosynthesis of the three?

(A) Pigments involved in photosynthesis (2 points)

Purple sulfur bacteria	
Blue-green bacteria	
Green algae	

(B) Location of

photosynthesis (2 points)

Purple sulfur bacteria	
Blue-green bacteria	
Green algae	

(C) Raw materials

used in photosynthesis (2 points)

Purple sulfur bacteria	
Blue-green bacteria	
Green algae	

(D) Products of photosynthesis (including byproducts) (3 points)

	Main products of photosynthesis (organic matter)	Byproducts of photosynthesis
Purple sulfur bacteria		
Blue-green bacteria		
Green algae		

3. Glucocorticoids (a hydrophobic molecule) and glutamate (an amino acid) are two types of hormones. Please explain how these two hormones induce responses in their target cells. **(8 points)**

V. Multiple Choice Questions (9 points in total)

- _____ 1. The activity of cyclin-dependent kinases (CDKs) can be increased under the following conditions.

Strong, except for **(3 points)**

- (A) High concentration of cyclin
- (B) Dephosphorylated by CDC25
- (C) High concentration of p53 protein
- (D) Degradation of p21 protein
- Phosphorylation of (E) cyclin B1

- _____ 2. Which of the following amino acids has an R group containing a hydroxyl group (-OH)? **(3 points)**

- (A) Glycine and proline
- (B) Serine and tyrosine
- (C) Leucine and Tryptophan
- (D) Aspartate and glutamine
- (E) Histidine and lysine

- _____ 3. As an example of covalent bonds, which of the following makes a significant contribution to maintaining the tertiary structure of proteins? **(3 points)**

- (A) Van der Waals forces
- (B) Hydrogen bonds
- (C) Disulfide bridges
- (D) Hydrophobic interaction
- (E) DNA cross-link

VI. Matching Questions

1. Please fill in the correct definitions (codes) for the proper nouns. **(8 points)**

proper nouns	definition
Apoptosis	
Cell necrosis	
Autophagy	
senescence	

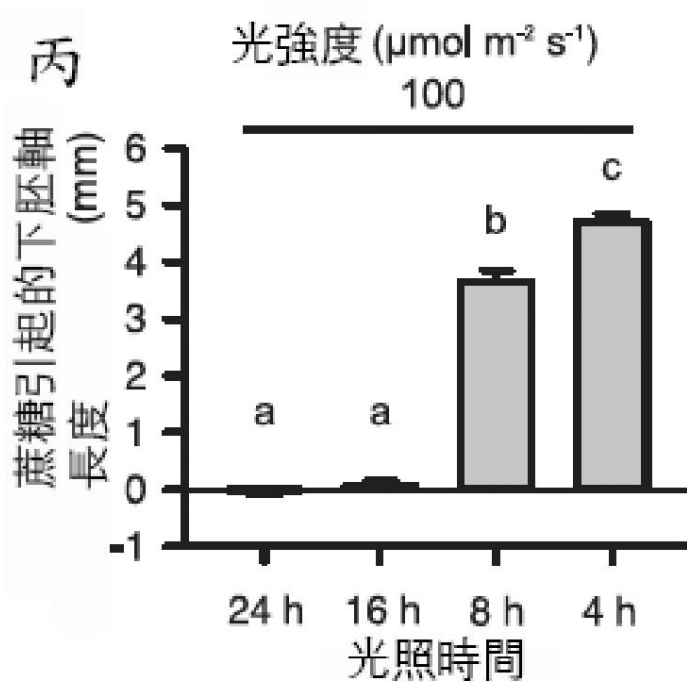
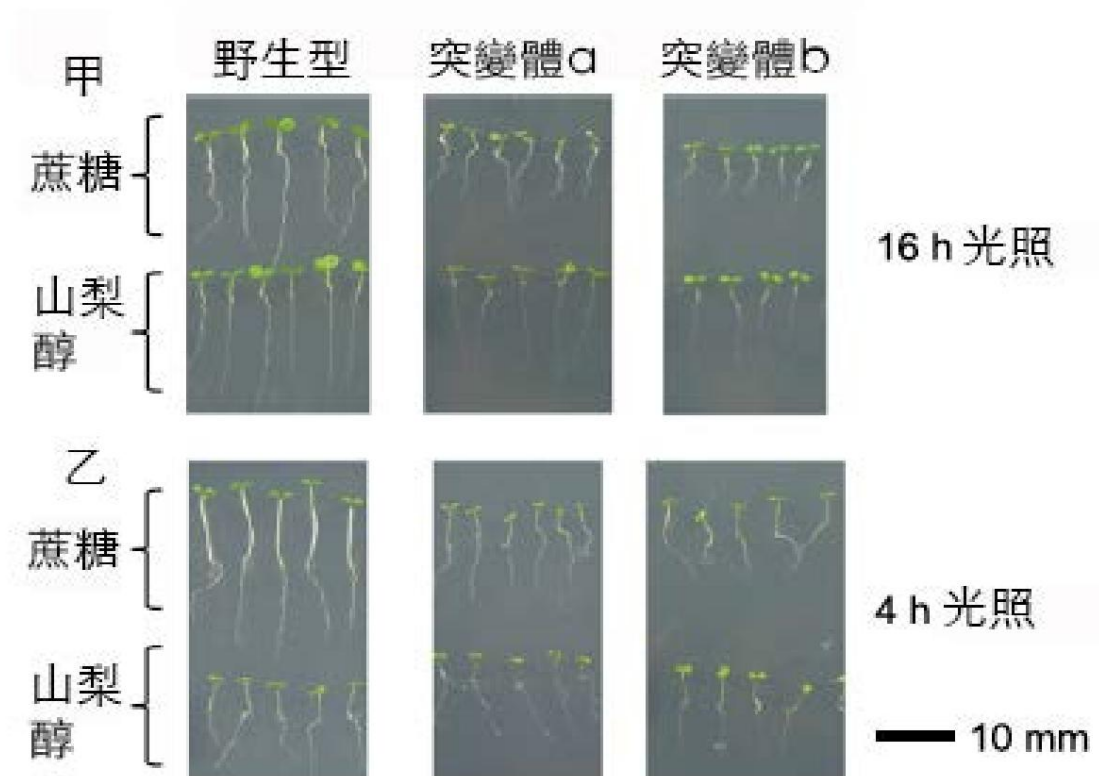
code name	definition
a.	Degradation of intracellular components via lysosomes
b.	Cell cycle arrest
c.	Premature cell death caused by external factors
d.	Programmed cell death Part of it is regulated by growth

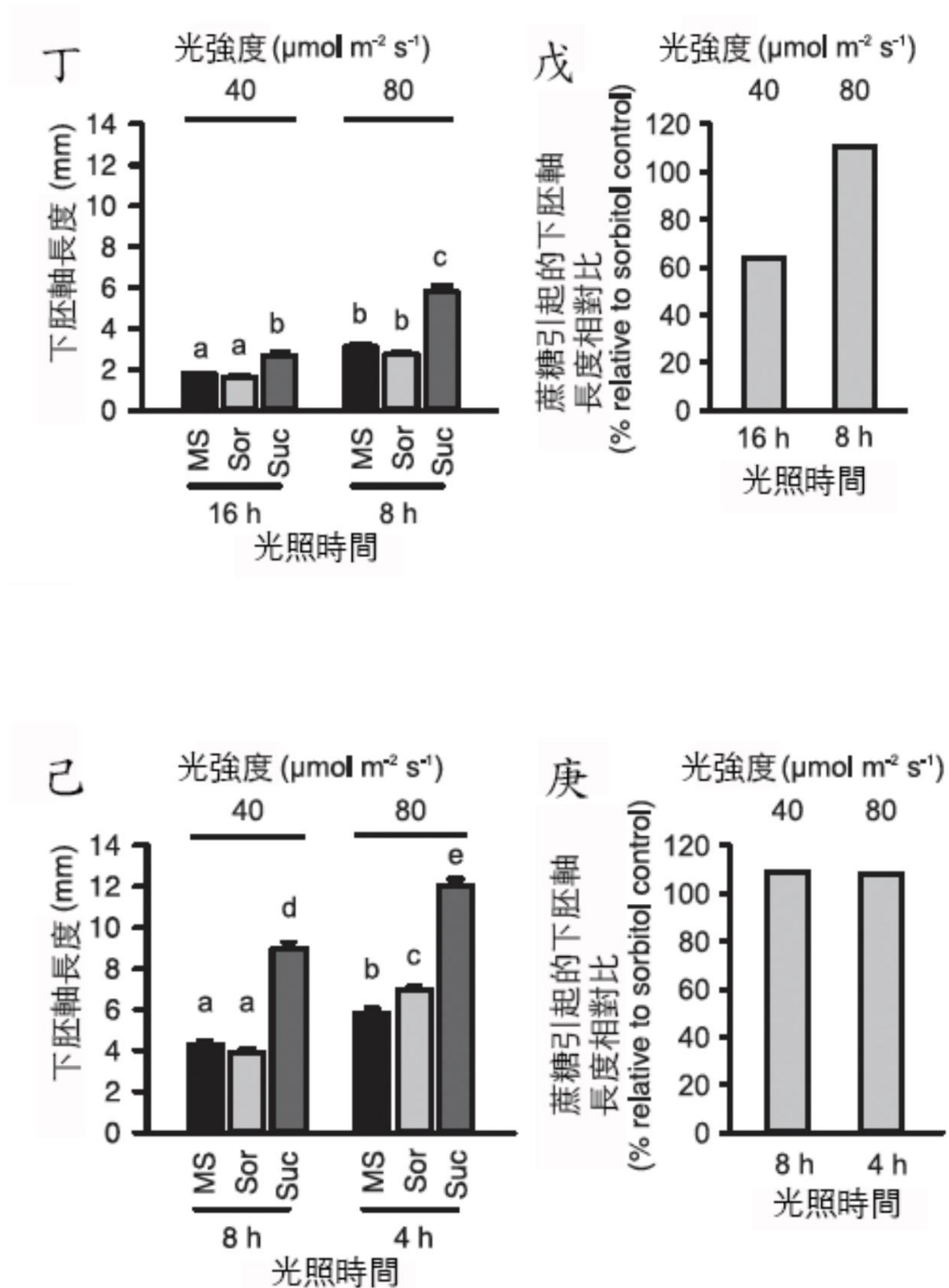
Part Two: (100 points)

A student is very interested in the effects of sugars and light on plant seedlings, and therefore uses Arabian mustard seedlings under light...

Experiments were conducted under different environmental conditions to observe the effects of adding different sugars on seedling growth. The

experimental results are as follows: Figures (A-G) show the relationship between sucrose (Suc) or sorbitol (Sor) and the effects of photoperiod and light intensity on the hypocotyl length of Arabian mustard seedlings.





Based on the graph of the experimental results, please answer the following questions:

1. Based on Figures A and B, which type of sugar has the most significant effect on the wild-type hypocotyl under light ? **(2 points)** Describe the relationship between this sugar and the duration of light exposure . (4 points) What is the most likely reason for this ? **(4 points)**

2. Based on Figures A and B, what are the responses of mutant a and mutant b to sucrose? Explain **(5 points)**.

3. Based on Figure C, describe the relationship between sucrose, light intensity, and light duration on the hypocotyl length **(5 points)**?

4. Based on Figure D, describe the relationship between these two sugars and the hypocotyl length regulated by light intensity and photoperiod duration **(5 points)**?

5. According to Figure E, if the light intensity increases to $120 \mu\text{mol m}^{-2} \text{s}^{-1}$, it will cause a relative increase or decrease in the hypocotyl ratio.

Low? (2 points) Explain why? (5 **points**)

6. Based on Figures F and D, describe the relationship between these two sugars and the hypocotyl length regulated by light intensity and photoperiod duration. (5 **points**)

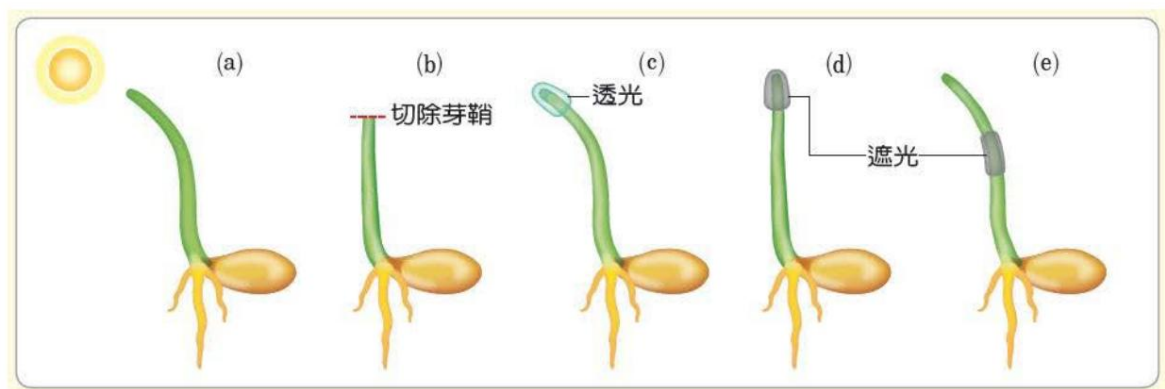
7. Based on Figures E and G, describe the relationship between sucrose and hypocotyl length regulated by light intensity and photoperiod. (5 **points**)

8. Based on diagrams A through G, summarize the effects of sucrose and light on hypocotyl growth. (8 **points**)

II. Based on previous research on the discovery of plant auxin, trace back and explore the experimental design and interpretation of the results of the original research. There are 4 sets of experiments below. The schematic diagram of each set of experiments shows the results of multiple sub-experiments in the research of that era. Please fill in the following in the table in order: (1) Question and hypothesis. (2) Experimental design: In the table of each set of experiments, write down the experimental group and control group of each sub-experiment, as well as the manipulated variable to be tested. (3) Interpretation of results: According to the diagram, write down the results of each sub-experiment, and summarize the total results (conclusion) of the set of experiments at the end of each table. The first table is an example of how to fill in the table. Please add rows and columns below each table to fully present the research results in the schematic diagram of each set of experiments.

1. In 1880, Darwin and his son noticed that the coleoptiles of a type of grass would grow towards the light, so they conducted experiments to explore this phenomenon.

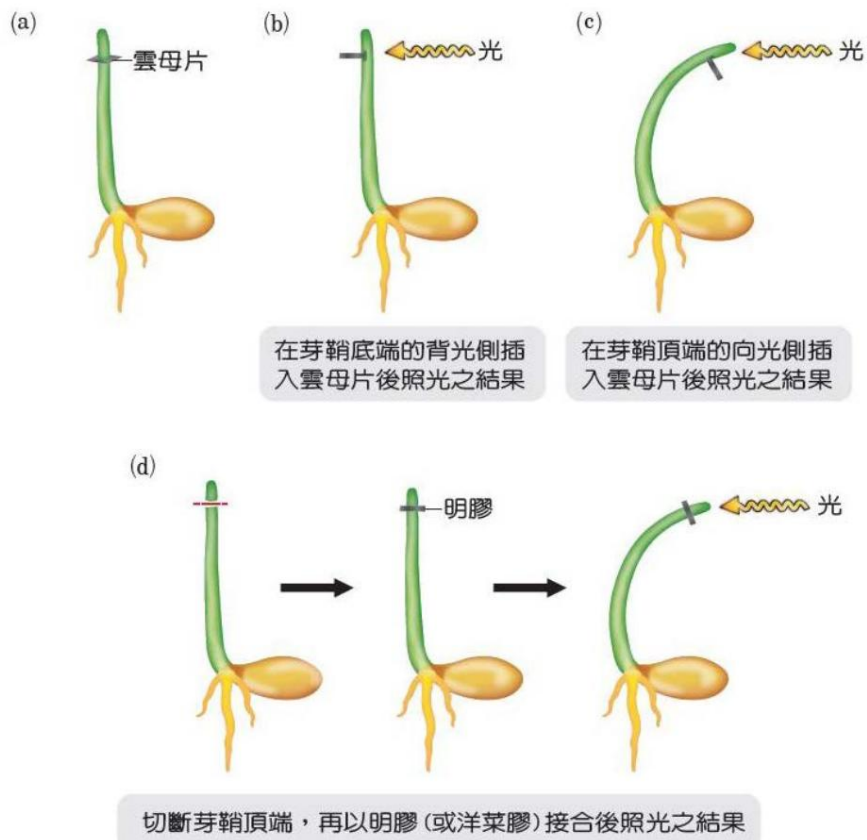
The results of multiple sub-experiments are shown in the figure below.



Fill in the form (10 points)

Question and Hypothesis		Question: Why do coleoptiles grow towards the light? Hypothesis: The tip of the coleoptile can sense side light.				
Experimental Design	Experimental Group	Control Group	Manipulation Variable	Light Treatment	Experimental Results	Experiment 1: Removal of the coleoptile tip
(b)			Complete coleoptile (a)	Presence or absence of coleoptile apex	Side-lit, apical coleoptiles will not grow towards the light.	
Experiment 2						
Experiment 3						
Results Summary		(in conclusion)				

2. In 1913, P. Boysen-Jensen (1883-1960) separated the tips of coleoptiles using mica sheets and gelatin, respectively. The results of several sub-experiments are shown in the figure below. (Note: The figure only shows the key results and does not present all experimental design combinations.)



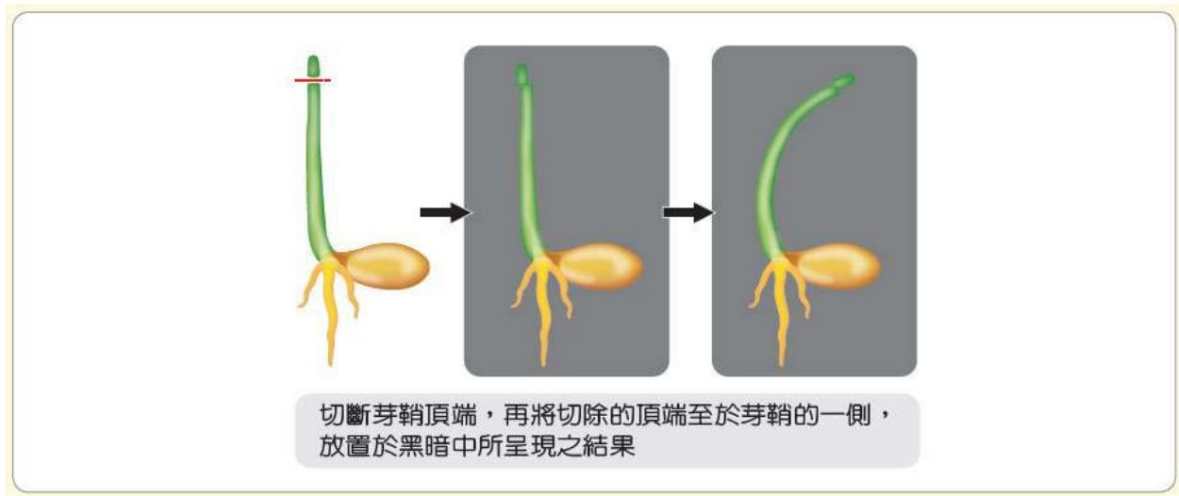
Fill in the table (15 points). You may add rows and columns as you wish to fully present the research results shown in the diagrams for each group of experiments.

Problems and Hypotheses:	hypothesis:				
Experimental Design, Experimental Group, Control Group, Variables, Illumination Treatment, Experimental Results					
Experiment 1					
Experiment 2					
Results Summary (in conclusion)					

3. In 1918, A. Paál (1889-1943) placed the severed coleoptile tip sideways and grew it in the dark, observing the growth results, as shown in the figure below.

(Note: The figure only shows the key results and does not present all the actual growth.)

(Experimental Design Team)



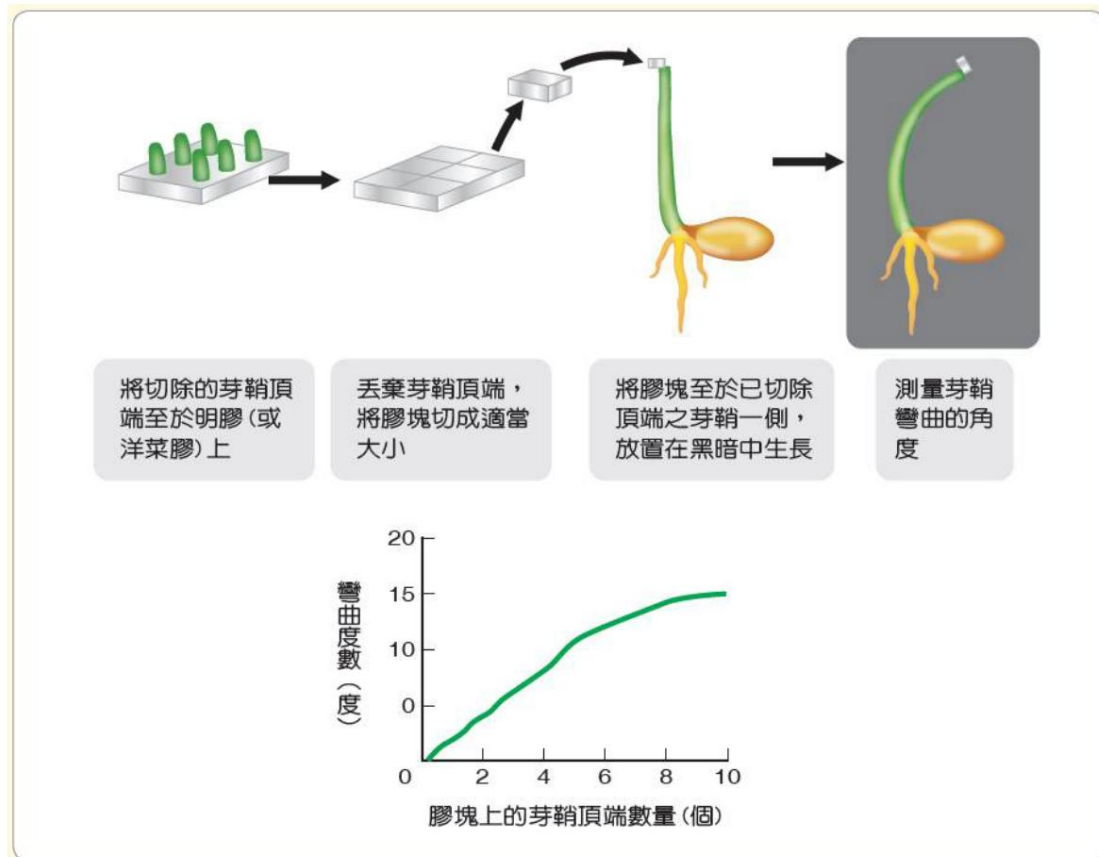
Fill in the table (**10 points**). You may add rows and columns as you wish to fully present the research results and hypothesis questions in each

experimental diagram:	hypothesis:				
Experimental Design, Experimental Group, Control Group, Variables, Illumination Treatment, Experimental Results					
Experiment 1					
Results Summary (in conclusion)					

4. In 1926, F.W. Went (1803-1990) placed the severed coleoptile tips on gelatin, then replaced the tips with gelatin and placed the coleoptiles on their sides, growing them in the dark to observe the bending of the coleoptiles; and used different amounts of...

The bending angle was measured at the tip of the coleoptile, and the results are shown in the figure below. (Note: Only the key results are shown in the figure.)

(Not all experimental design groups are fully presented)



Fill in the table (15 points). You may add rows and columns as you wish to fully present the research results shown in the diagrams for each group of experiments.

Problems and Hypotheses:	hypothesis:				
Experimental Design, Experimental Group, Control Group, Variables, Illumination Treatment	Experimental Results				
Experiment 1					
Results Summary (in conclusion)					

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Theoretical Assessment Test Questions

Part Three

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Part Three: (100 points)

I. Background Introduction:

The myelin sheath in the central nervous system contains proteins such as myelin-associated glycoprotein (MAG), which inhibit neuronal regeneration when neurons are damaged. Previous research has found that increasing the activity of the cyclic adenosine monophosphate (cAMP) signaling pathway within nerve cells using the drug dbcAMP can effectively reverse the inhibitory effect of central nervous system myelin sheath on nerve growth; in addition, there is also evidence showing...

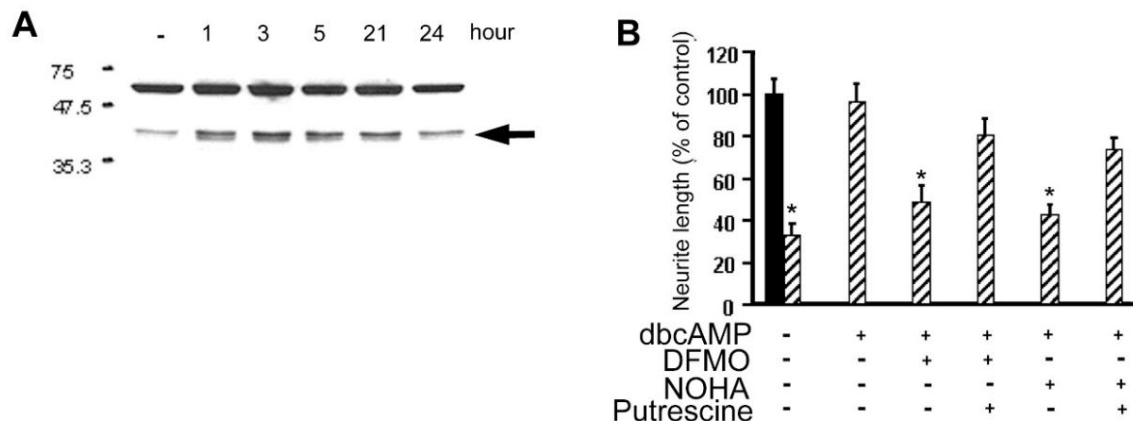
In an in vitro cell culture environment, the additional administration of putrescine (a polyamine that can be produced by stimulating cells) to nerve cells can reverse the myelin's inhibitory effect on nerve growth; however, the mechanism by which these substances reverse nerve growth remains unclear. The following experiment uses two types of central nervous system cells—dorsal root ganglion neurons (DRGs) and cerebral granule neurons (CNGs)—as experimental models to further explore their mechanisms of action.

Arginase-I (Arg-I) is a key enzyme in the intracellular synthesis of the polyamine compound putrescine; stimulating its activity can promote the production of more putrescine within the cell. Furthermore, previous research has shown that the intracellular protein mechanism Cdk5 (cyclin-dependent kinase 5) is involved in the rearrangement of the cytoskeleton during nerve cell growth.

The activity of the Cdk5 protein is regulated by another intracellular activating protein, p35.

Experimental results:

(Figure 1) (A) After nerve cells were treated with dbcAMP for a period of time (as shown in the upper part of the figure), the cell products were dissolved and subjected to protein electrophoresis. The amount of Arg-I present in the electrophoretic gel was detected by antibody. (B) DRG neurons were cultured in culture dishes with central nervous system myelin layer at the bottom (horizontal bar graph) or without central nervous system myelin layer (black bar graph). Various experimental combinations of dbcAMP, DFMO (Arg-I inhibitor), NOHA (Arg-I inhibitor), or putrescine were designed to be administered alone or simultaneously. The changes in nerve growth length were observed. The asterisk indicates that the comparison is statistically significant.



(Figure 2) CNG neurons (A) or DRG neurons (B) after being administered DRB (transcriptional neuron) alone or simultaneously.

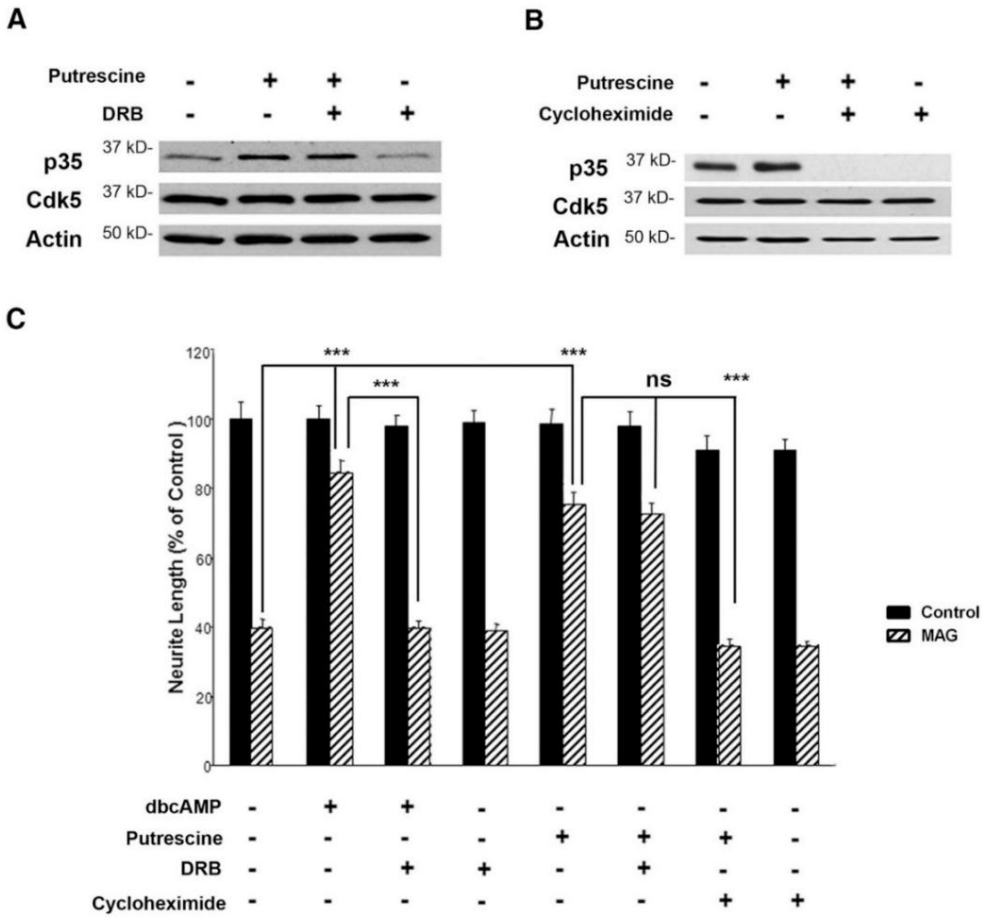
Design various experimental combinations using inhibitors, cycloheximide (a translational inhibitor) , or putrescine.

Subsequently, the cell products were dissolved and subjected to protein electrophoresis, and antibodies were used to detect the presence of p35 or Cdk5 on the electrophoretic gel, respectively.

The amount was [not specified], with the Actin group serving as the control group. (C) CNS neurons were cultured in regular culture dishes (control group, black bars).

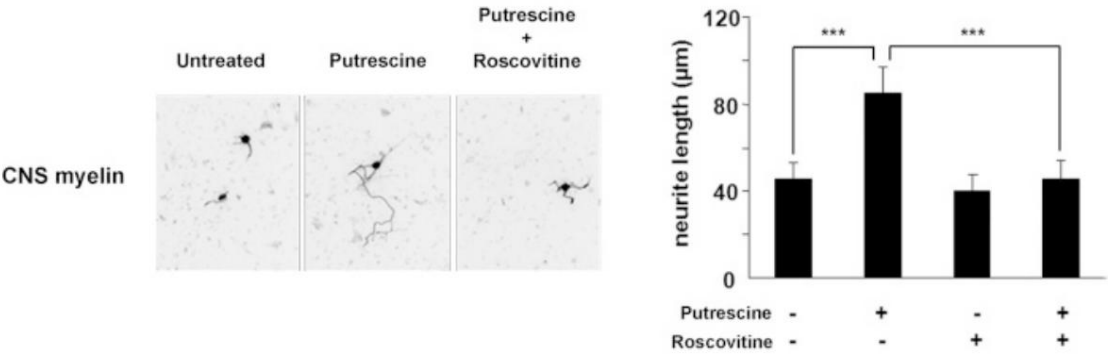
Alternatively, in a culture dish containing MAG (bar graph), changes in nerve cell growth length can be observed under conditions where dbcAMP, putrescine, DRB, or cycloheximide are administered simultaneously. An asterisk indicates a comparative effect.

It has statistical significance, while ns represents that it does not have statistical significance.



(Figure 3) The growth of DRG neurons after culturing them in a medium coated with central nervous system myelin was observed, with and without drug treatment, and in the presence of putrescine and roscovitine (Cdk5 inhibitors) (left) . The right figure shows the statistical results.

An asterisk indicates that the comparison has statistical significance.



1. Please synthesize all experimental results to explain how cAMP reverses the inhibition of neuronal cell growth by central myelin sheath substances.

Mechanism. (30 points)

II. Afen is a middle-aged married woman with three children. Recently, she has been experiencing fatigue, chest pain, palpitations, shortness of breath when squatting down to pick things up, difficulty climbing half a flight of stairs, severe swelling in her legs, and has even experienced her lips turning blue and suddenly fainting. After seeking medical attention, the doctor conducted several tests, including an electrocardiogram, echocardiogram, chest X-ray, pulmonary function tests, arterial blood gas analysis, and right heart catheterization. Afen was diagnosed with "pulmonary hypertension." Please answer the following questions based on your knowledge of physiology.

1. Many factors can lead to pulmonary hypertension. Please explain why the following factors might induce pulmonary hypertension.

(1) Long-term exposure to high-altitude environments. (5

points) (2) Hypertension of the portal vein system. (10 **points)**

2. In the early stages, there were few treatment options for pulmonary hypertension, and patients often had to rely solely on lung transplantation or heart-lung transplantation.

However, thanks to the efforts of scientists, many drugs have been developed for the treatment of pulmonary hypertension. Please specify the following drugs.

Can this substance be used to treat pulmonary hypertension? Explain its mechanism of action clearly. Only correct answers will receive points; incorrect answers will result in deductions.

0.5 points, deducting up to 0 points for this question. (15 **points**) (a)

Calcium channel blocker (b)

Prostacyclin (c) Endothelin receptor antagonist (d)

Diuretic (e) Phosphodiesterase type 5

inhibitor

III. Please answer the following questions related to human evolution.

1. Some scholars believe that *Homo erectus* is the direct ancestor of Neanderthals, Denisovans, and *Homo sapiens*. Which speciation process—*anagenesis* or *cladogenesis*—does this view align with? Why? Please provide a diagram or explanation. (10 **points**)

2. Please explain the morphological and physiological costs that *Hominii* animals paid for bipedalism.

How can such a cost be compensated? (10 **points**)

IV. Many animals possess photonic crystals on their body surface. Please explain the benefits of having photonic crystals for animals.

What kind of functions? (10 **points**)

V. What effects might genome duplication have? (10 **points**)

2019 Republic of China Biology Olympiad National Team

Selection Camp

Theoretical Assessment Test Questions

Part Four

Student Code



Precautions:

1. This section of the test consists of 3 pages (including the cover) and is worth 100 points.
2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please make sure that the cover of the exam paper matches your exam number.
4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part Four: (100 points)

1. The yeast translation initiation factor Tif3 gene encodes an RNA-binding protein. Its N-terminus contains an RNA recognition motif (RRM), followed by seven repeat sequences (numbered 1 to 7) rich in basic and acidic amino acids. The C-terminal sequence lacks a clearly known functional region. Previous research by Xiaoying has shown that Tif3 gene deletion causes a temperature-sensitive growth defect phenotype, which can be compensated for by exogenously transferring wild-type Tif3 expression plasmid DNA.

The study also revealed the impairment of Tif3 gene function and that the Tif3 protein possesses at least two RNA-binding domains and can catalyze RNA.

RNA strand exchange. Xiaoying wants to further design experiments to investigate the effects of Tif3 functional region defect phenotypic expression plasmids on the growth of Tif3 gene-deficient yeast and its in vitro translation activity. The symbol + represents the presence of...

The symbol - represents nothing. The analysis results are shown in the figure below.

Tif3蛋白		37°C生長狀況	轉譯活性(%)	RNA股交換
N	RRM 1 2 3 4 5 6 7 C	++	100	+++
N	RRM 1 2 3	++	43	++
N	RRM 1	-	0	-
	1 2 3 4 5 6 7 C	+	0	+

Based on the above experimental results, please explain and deduce the following questions:

- (1) Please deduce and explain what the main functional region of the Tif3 protein is involved in translation. (20 points)

- (2) How is the RRM structure related to yeast growth? (20 points)

(3) Please explain why the RNA strand exchange activity of the Tif3 protein is required for the initiation of protein translation. **(40 points)**

(4) Please explain the possible role of the RRM in the Tif3 protein's role in protein translation initiation. **(20 points)**

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Theoretical Assessment Test Questions

Part Five

Student Code



Precautions:

1. This section of the test consists of 8 pages (including the cover) and is worth 100 points.
2. Please check for missing pages. If any pages are missing, please inform the test administrator immediately.
3. When answering the questions, please check if the cover of the test paper matches your test number.

4. Please answer the questions on the test paper using a blue (black) pen, and please keep the test paper clean.
5. You may write your answers on the back of the questions, but please indicate the question number.

Part Five: (100 points)

I. Life tables track and record mortality rates at different age stages within a population, and are a crucial tool for studying population ecology.

Required data. Suppose an ecologist tracks and records individuals of the same population of two species, A and B, from birth...

Mortality rates at different age stages until death.

1. Based on the mortality rates at different age stages, calculate the number of surviving individuals of species A and B at different age stages.

Both species A and B divide age into 11 stages (0-9, 10-19,90-99, 100+), and the mortality rate is defined as a certain...

The percentage of individuals who die within an age range from beginning to end; the number of surviving individuals is defined as the percentage of individuals who survive from birth to the end of a certain age range.

Starting from the age stage, how many individuals are still alive? Assume the survival rate of species A and B during the first age stage (0-9 years) is...

The number of surviving individuals was 100 for all age groups. The number of surviving individuals for other age groups was recorded to one decimal place.

(Round to the nearest whole number). (1 point per blank, 20 points in total)

Species A

Age group	mortality rate	Number of surviving individuals
0-9	0.6	100
10-19	0.5	)
20-29	0.05	)
30-39	0.05	)
40-49	0.05	)
50-59	0.2	)
60-69	0.4	)
70-79	0.3	)
80-89	0.3	)
90-99	0.8	)
100+	1	(((((

Species B

Age group	mortality rate	Number of surviving individuals
0-9	0.1	100
10-19	0.01	)
20-29	0.05	)
30-39	0.05	)
40-49	0.05	)
50-59	0.2	)
60-69	0.3	)
70-79	0.5	)
80-89	0.8	)
90-99	0.8	)
100+	1	(((((

2

Part Five

2019 Selection Camp Theory Student Code:

2. Represent the relationship between the age stages (X-axis) and the number of surviving individuals (Y-axis) of species A and B using line graphs. (5 points for each graph , **10** points in total. Graphs must be complete and clear.)

Species A

Species B

3. Answer the following statements as either correct (O) or incorrect (X). (2 points each , **4 points in total**)

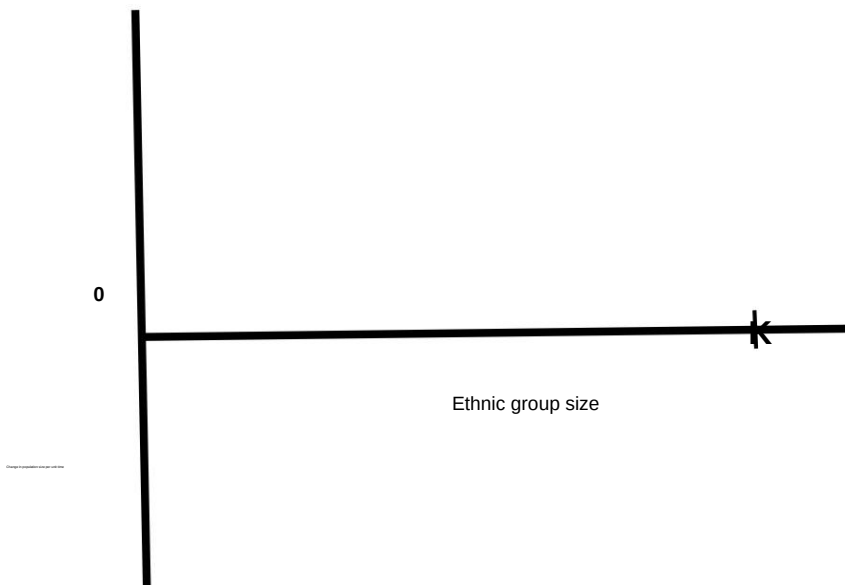
(1) () Based on the life tables of species A and B above, the population size of species A may be smaller than that of species B.

(2) () Species B may exhibit parental care behavior compared to species A.

II. In nature, the increase or decrease in the population size depends on the population size relative to carrying capacity.

Carrying capacity is either high or low. It represents the number of populations an environment can support. When the population size is below carrying capacity, the population will increase, especially when it is significantly below carrying capacity, the increase will be faster. Conversely, when the population size is above carrying capacity, the population will decrease. However, some species face the so-called Allee effect. As mentioned earlier, under normal circumstances, when the population size is significantly below carrying capacity, the population size increases faster. But when the Allee effect occurs, it means that when the population size is significantly below carrying capacity, the population size actually decreases. Because many endangered species have populations far below carrying capacity, conservationists are quite concerned about whether the Allee effect will occur, meaning that already scarce species may become even scarcer.

1. Plot the relationship between population size and population size change per unit time for the populations without and with the Allee effect in the following graphs (draw them on the same graph, but clearly label them), where K represents carrying capacity. (5 points for each graph, **10 points in total**)



2. Give two scenarios that could cause the Allee effect (3 points each, **6 points in total**; only the first two answers will be counted).

point)

III. Experiments on Bird Feeding Preference Behaviors

To understand the feeding preferences of fruit-eating birds, researchers kept four common fruit-eating bird species found at low altitudes in Taiwan in captivity: three Light-vented Bulbuls (average beak base width 10.3 mm), four Red-billed Bulbuls (average beak base width 11.3 mm), three Parakeets (average beak base width 16.2 mm), and five Tree Magpies (average beak base width 16.5 mm). They also provided artificial masks of different colors and sizes.

The group was used to test the size and color of the food preferred by these four bird species.

The beak base width was measured using a vernier caliper, from the leftmost to the rightmost side of the inner beak base of each experimental bird. All experimental birds were from the wild. Each bird was individually housed in a 76×46×46 cm cage with a white bottom.

In the morning, the birds are given fresh and plentiful feed, water, and a small piece of papaya or banana to meet their basic nutritional needs.

All bird feeding preference experiments were conducted in the morning in a closed indoor environment. Before the experiments, the bird feed bowls were removed and the birds were allowed to roam freely.

The birds stopped eating for two hours before the experiment began. Each bird was tested only once per day.

In a food color preference experiment, dough was treated with five colors: red, black, yellow, brown, and green, and all were tested together. The dough pieces were all 5 mm in diameter and weighed approximately 0.2 g each. In this experiment, the dough was identical in composition and other characteristics except for color. The five food bowls were placed randomly. All 15 experimental birds underwent repeated testing.

Ten tests were conducted, for a total of 150 tests.

In the food size preference experiment, the dough was prepared in three diameter grades: 5 mm (approximately 0.2 g), 9 mm (approximately 0.8 g), and 12 mm (approximately 1.5 g). All dough samples were red in color and identical in composition and other characteristics except for diameter grade. The three food bowls were placed randomly, and the experiment was repeated 12 times for each of the 15 experimental birds.

The experiment involved 180 tests.

After the dough was placed in the cage, researchers observed the birds' behavior and removed all the dough 40 minutes later.

The researchers calculated the following data based on behavioral observations and the remaining dough weight: [The researchers then analyzed the dough bowl and calculated the remaining weight.]

(1) Priority Selection Index: Scored according to the order in which birds choose colors. In the color preference experiment, the highest priority selection index is given to the birds.

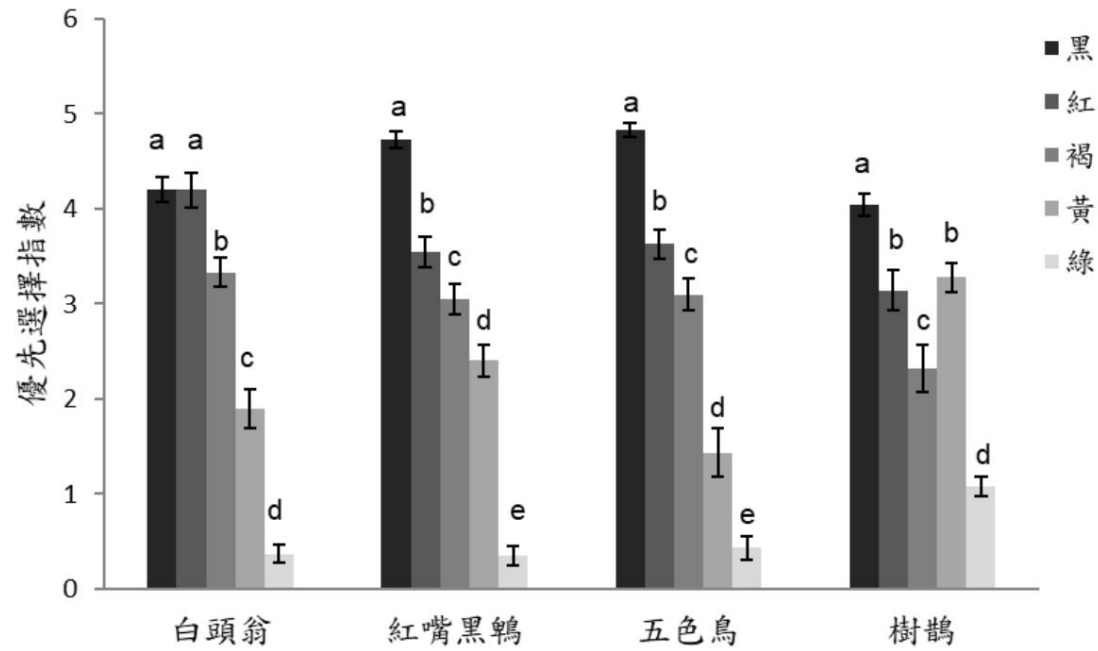
The color is awarded 5 points, the next 4 points, and so on. The last selected color receives 1 point, and the unselected color receives 0 points.

There are six levels in total. In the size preference experiment, the diameter class most preferred by birds was given a score of 3, followed by 2 points,

The last choice is worth 1 point, and the last choice is worth 0 points. There are four levels in total.

(2) Feed weight: The amount of dough put in minus the amount of dough left, multiplied by the average weight of a single piece, represents the weight of food that the bird attempted to eat (including food that fell on the tray).

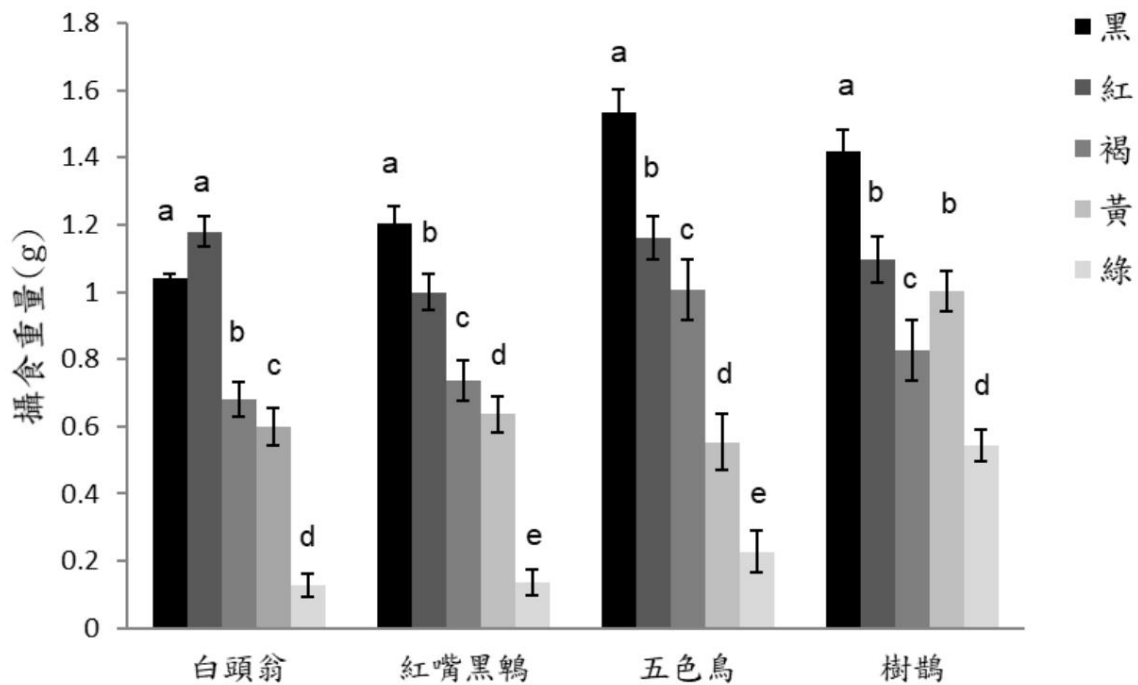
The experimental results are presented in the following four figures.



(Figure 1) Preferred selection indices of five colored doughs for the Light-vented Bulbul, Red-billed Bulbul, Five-colored Bird, and Tree Magpie. The letters a, b, c, etc.

in the figure represent whether there are significant differences in the preferred selection indices of different colored doughs among the bird species. If the same letter

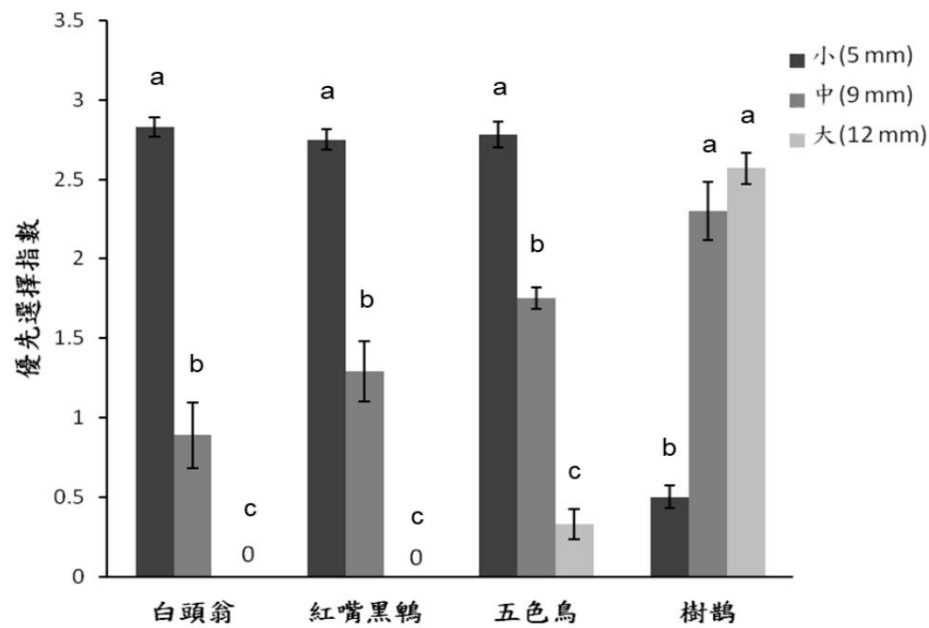
is found between treatments, it indicates that there are no significant differences between them.



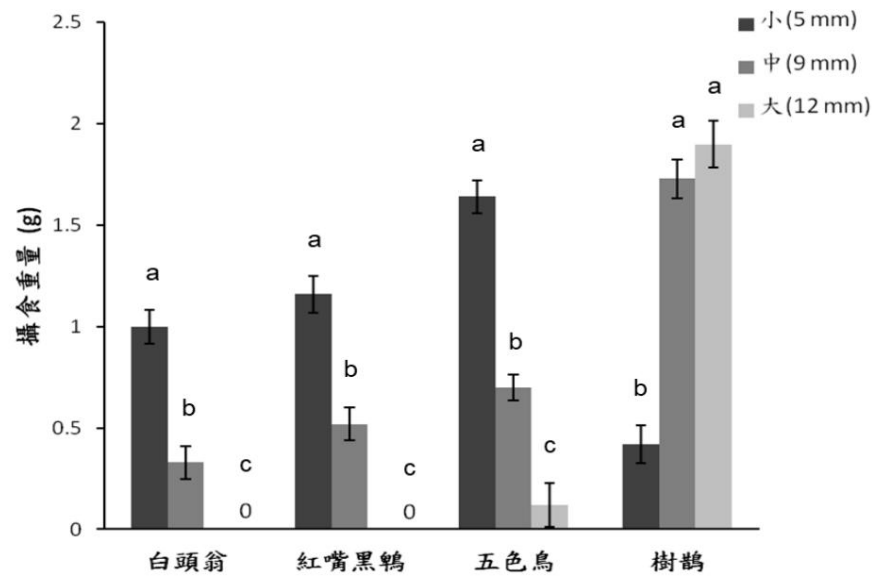
(Figure 2) Feed weight of five colored doughs for the Light-vented Bulbul, Red-billed Bulbul, Five-colored Bird, and Tree Magpie. The letters a, b, c, etc. in the figure

represent whether there is a significant difference in the feed weight of different colored doughs for each bird species. If the same letter is present between treatments,

it means that there is no significant difference between them.



(Figure 3) Preferred selection indices of three dough sizes for the Light-vented Bulbul, Red-billed Bulbul, Common Bluetail, and Tree Magpie. The letters a, b, c, etc. in the figure represent whether there are significant differences in the preferred selection indices of different dough sizes among the bird species. If the same letter is present in the treatments, it indicates that there are no significant differences between them.



(Figure 4) Feed weight of three diameter doughs for the Light-vented Bulbul, Red-billed Bulbul, Common Bluetail, and Tree Magpie. The letters a, b, c, etc. in the figure represent whether there are significant differences in the feed weight of different dough diameters for each bird species. If the same letter is present between treatments, it indicates that there are no significant differences between them.

IV. True or False Questions (50 points in total, 6 points will be deducted for each wrong answer , up to a maximum of 0 points for this section)

Which of the following statements is true? If true, please fill in "O" before the question number; if false, please fill in "X".

____ 1. For plants, seeds that fall under the mother tree have a lower survival rate than seeds that are far away from the mother tree because they have to compete with the mother tree.

____ 2. Co-evolution between plant and bird seed dispersers will prompt plants to develop specific seed dispersers.

Fruit characteristics.

____ 3. The experimental results all support the hypothesis that "all four bird species prefer black food".

____ 4. The experimental results all support the hypothesis that "all four bird species prefer food with smaller diameters".

____ 5. The experimental results all support the view that "these four bird species have varying degrees of preference for different food colors and diameter classes, with a preference index and..."

The results were consistent with the statement that "the weight of food consumed was consistent with the results of this statement."

____ 6. Scientists have proposed the "gap limitation hypothesis," which suggests that the base width of a bird's beak is limited.

The larger the fruit, the larger its diameter can be swallowed. If the fruit is larger than the width of the bird's beak base, it must be pecked.

They obtain energy by eating fruit. The results of this experiment all support the beak base width limitation hypothesis.

____ 7. The optimal foraging theory posits that animals will employ methods that expend the least amount of energy when foraging.

The strategy to maximize benefits. Because birds expend less time and energy eating by swallowing than by pecking.

Based on the theory of optimal foraging strategies, we can deduce that "within the range that a bird's beak can swallow, it will choose..."

Statements such as "choose the largest food size to improve foraging efficiency" are common.

____ 8. The experimental results all support the view that "within the range that a bird's beak can swallow, birds will choose the largest food size to improve their foraging ability."

The "efficiency" hypothesis.

____ 9. Because green fruit usually indicates that the fruit is unripe and contains toxic secondary compounds.

(compounds), so birds avoid eating green fruits. This experiment supports this claim.

____ 10. The results of this experiment all support the view that "compared to the green branches and leaves of plants, green fruits have little contrast with the background, therefore birds..."

The hypothesis is that "humans are less likely to eat green fruits."

____ 11. If laboratory animals are individuals that have been cage-raised for many years and fed artificial dough for an extended period, under the same experimental procedures,

Their preference for dough color should be consistent with that of this experiment.

____ 12. If laboratory animals are individuals that have been cage-raised for many years and fed artificial dough for an extended period, under the same experimental procedures,

Their preference for dough diameter should be consistent with that of this experiment.